

📌 12-MONTH DATA ENGINEERING ROADMAP

From Zero to Data Engineer | Month-by-Month | Skill-by-Skill

Duration: 12 Months

Time Required: 25-30 hours/week

Cost: ₹0 (100% Free Resources)

Target Role: Data Engineer at ₹12-18 LPA

📌 MONTH 1-2: SQL MASTERY

Skills to Learn:

- SQL Basics (SELECT, WHERE, GROUP BY, ORDER BY)
- Joins (INNER, LEFT, RIGHT, FULL, CROSS)
- Subqueries and CTEs (WITH clause)
- Window Functions (ROW_NUMBER, RANK, LAG, LEAD)
- Query Optimization (EXPLAIN, Indexes)
- Database Design (Normalization, ERD)

Free Resources:

Interactive Tutorials:

- Mode SQL Tutorial: <https://mode.com/sql-tutorial/>
- SQLZoo Interactive: <https://sqlzoo.net/>
- W3Schools SQL: <https://www.w3schools.com/sql/>
- PostgreSQL Tutorial: <https://www.postgresqltutorial.com/>

Video Courses:

- Alex The Analyst - SQL Full Course (4 hours): <https://www.youtube.com/watch?v=HXV3zeQKqGY>
- freeCodeCamp - SQL Course (4 hours): <https://www.youtube.com/watch?v=HXV3zeQKqGY>
- Programming with Mosh - SQL (3 hours): https://www.youtube.com/watch?v=7S_tz1z_5bA

Practice Platforms:

- LeetCode Database Problems: <https://leetcode.com/problemset/database/>
- HackerRank SQL: <https://www.hackerrank.com/domains/sql>
- Codecademy SQL: <https://www.codecademy.com/learn/learn-sql>

Documentation:

- PostgreSQL Docs: <https://www.postgresql.org/docs/>
- MySQL Docs: <https://dev.mysql.com/doc/>
- Use The Index, Luke (Performance): <https://use-the-index-luke.com/>

Week-by-Week Breakdown:

Week 1-2: SQL Basics

- Learn: SELECT, WHERE, ORDER BY, LIMIT, DISTINCT
- Learn: Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- Learn: GROUP BY and HAVING
- Practice: Solve 20 easy problems on LeetCode

Week 3-4: Joins and Subqueries

- Learn: All JOIN types with examples
- Learn: Subqueries in SELECT, WHERE, FROM
- Learn: Common Table Expressions (CTEs)
- Practice: Solve 20 medium problems on LeetCode

Week 5-6: Window Functions

- Learn: ROW_NUMBER(), RANK(), DENSE_RANK()
- Learn: LAG(), LEAD(), FIRST_VALUE(), LAST_VALUE()
- Learn: PARTITION BY and frame clauses
- Practice: Solve 15 window function problems

Week 7-8: Optimization & Design

- Learn: EXPLAIN and query execution plans
- Learn: Indexes (B-tree, Hash)
- Learn: Database normalization (1NF, 2NF, 3NF)
- Learn: Transactions and ACID properties
- Practice: Optimize slow queries

Projects:

Project 1: E-Commerce Database Design

- Design schema for Users, Products, Orders, Reviews
- Implement in PostgreSQL with proper normalization
- Create indexes on frequently queried columns
- Write 20 analytical queries
- Document with ERD diagram

Project 2: Sales Data Analysis

- Download Superstore dataset: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>
- Write 15+ complex analytical queries
- Use window functions for rankings
- Use CTEs for complex logic
- Document insights found

Project 3: Query Optimization Challenge

- Create large test dataset (1M+ rows)
- Write 10 slow queries
- Use EXPLAIN to analyze
- Optimize with indexes and query rewrites
- Document performance improvements

Tools to Install:

- PostgreSQL: <https://www.postgresql.org/download/>
 - DBeaver (DB Client): <https://dbeaver.io/>
 - pgAdmin: <https://www.pgadmin.org/>
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📅 MONTH 3-4: PYTHON FOR DATA ENGINEERING

Skills to Learn:

- Python Fundamentals (variables, loops, functions)
- Data Structures (lists, dicts, tuples, sets)
- Pandas (DataFrames, data manipulation)
- NumPy (arrays, numerical operations)
- File Handling (CSV, JSON, Excel, Parquet)
- API Integration (requests library)
- Database Connectivity (psycopg2, SQLAlchemy)
- Error Handling and Logging

Free Resources:

Python Basics:

- Python.org Official Tutorial: <https://docs.python.org/3/tutorial/>
- Learn Python Interactive: <https://www.learnpython.org/>
- W3Schools Python: <https://www.w3schools.com/python/>
- Real Python Tutorials: <https://realpython.com/>
- Python Cheat Sheet: <https://www.pythoncheatsheet.org/>

Video Courses:

- Programming with Mosh - Python (6 hours): https://www.youtube.com/watch?v=_uQrJ0TkZlc
- freeCodeCamp - Python (4 hours): <https://www.youtube.com/watch?v=rfscVS0vtbw>
- Corey Schafer - Python Tutorials: <https://www.youtube.com/c/Coreyms>

Pandas & NumPy:

- Kaggle Python Course: <https://www.kaggle.com/learn/python>
- Kaggle Pandas Course: <https://www.kaggle.com/learn/pandas>
- Keith Galli - Pandas Tutorial: <https://www.youtube.com/watch?v=vmEHCJofslg>
- NumPy Tutorial: <https://www.youtube.com/watch?v=QUT1VHiLmml>
- Pandas Documentation: <https://pandas.pydata.org/docs/>

Practice:

- 100 Pandas Puzzles: <https://github.com/ajcr/100-pandas-puzzles>
- Pandas Exercises: https://github.com/guipsamora/pandas_exercises
- HackerRank Python: <https://www.hackerrank.com/domains/python>

Week-by-Week Breakdown:

Week 1-2: Python Fundamentals

- Learn: Variables, data types, operators
- Learn: Control flow (if/else, for, while)
- Learn: Functions, lambda expressions
- Learn: Lists, dictionaries, tuples, sets
- Practice: Solve 30 HackerRank Python problems

Week 3-4: Pandas & NumPy

- Learn: Pandas DataFrames and Series
- Learn: Reading CSV, Excel, JSON files
- Learn: Filtering, selecting, grouping data
- Learn: Merging, joining, concatenating
- Learn: Handling missing data
- Practice: Complete Kaggle Pandas course
- Practice: Solve 20 Pandas exercises

Week 5-6: File Handling & APIs

- Learn: CSV, JSON, Excel, XML file handling
- Learn: requests library for API calls
- Learn: Parsing JSON responses
- Learn: Error handling (try/except)
- Learn: Logging module

- Practice: Build API extraction scripts

Week 7-8: Database Integration

- Learn: psycopg2 for PostgreSQL
- Learn: SQLAlchemy ORM basics
- Learn: Executing SQL from Python
- Learn: Bulk data operations
- Learn: Building modular ETL scripts
- Practice: Build complete ETL pipelines

Projects:

Project 4: CSV/JSON Data Processing Pipeline

- Extract data from multiple CSV and JSON files
- Clean data (handle nulls, duplicates, data types)
- Transform data (aggregate, merge, calculate fields)
- Validate data quality
- Load to PostgreSQL database
- Add comprehensive logging
- Create config file for parameters

Project 5: API Data Extraction Pipeline

- Choose free API (OpenWeather, GitHub, NewsAPI)
- Build API client with error handling
- Extract data with rate limiting
- Transform and structure data
- Load to database and CSV
- Schedule with cron job
- Add retry logic for failures

Project 6: Excel to Database ETL

- Read multiple Excel sheets
- Validate schema and data types
- Clean data (remove nulls, duplicates)
- Transform data (unpivot, create derived columns)
- Load to PostgreSQL with proper data types
- Add data quality checks
- Document the pipeline

Tools to Install:

- Python 3.9+: <https://www.python.org/downloads/>
- Jupyter Notebook: <https://jupyter.org/install>
- VS Code: <https://code.visualstudio.com/>

📌 MONTH 5-6: APACHE SPARK & BIG DATA

Skills to Learn:

- Big Data Concepts (Distributed computing, HDFS)
- Spark Architecture (Driver, Executors, Cluster Manager)
- RDDs (Resilient Distributed Datasets)
- Spark DataFrames and Datasets
- Spark SQL
- PySpark (Python API for Spark)
- Transformations and Actions
- Partitioning and Bucketing
- UDFs (User Defined Functions)
- Window Functions in Spark
- Reading/Writing: Parquet, Avro, ORC
- Performance Optimization

Free Resources:

Learning Materials:

- Spark Official Documentation: <https://spark.apache.org/docs/latest/>
- Spark by Examples: <https://sparkbyexamples.com/>
- PySpark Tutorial: <https://spark.apache.org/docs/latest/api/python/>
- Databricks Academy (Free): <https://www.databricks.com/learn/training/home>

Video Courses:

- Spark Tutorial for Beginners: <https://www.youtube.com/watch?v=VlgqQf7vJ0Q>
- PySpark Full Course: https://www.youtube.com/watch?v=_C8kWso4ne4
- Databricks Spark Tutorial: <https://www.youtube.com/watch?v=XrpSRCwISdk>

Free Book:

- Learning Spark 2.0 (PDF): <https://pages.databricks.com/rs/094-YMS-629/images/LearningSpark2.0.pdf>

Practice Platform:

- Databricks Community Edition (Free): <https://community.cloud.databricks.com/>

Week-by-Week Breakdown:

Week 1-2: Spark Fundamentals

- Learn: Spark architecture and ecosystem
- Learn: RDDs - transformations and actions
- Learn: Spark DataFrames basics
- Learn: Reading and writing data
- Practice: Run examples on Databricks Community

Week 3-4: Spark SQL & DataFrames

- Learn: Spark SQL queries
- Learn: DataFrame operations (select, filter, groupBy)
- Learn: Joins in Spark
- Learn: Aggregate functions
- Practice: Process datasets on Databricks

Week 5-6: Advanced PySpark

- Learn: Window functions in Spark
- Learn: User Defined Functions (UDFs)
- Learn: Partitioning strategies
- Learn: Broadcast joins
- Learn: Performance tuning (caching, persist)
- Practice: Optimize Spark jobs

Projects:

Project 7: Large Dataset Processing

- Download 100MB+ dataset from Kaggle
- Load into Spark DataFrame on Databricks
- Perform transformations (filter, select, groupBy, join)
- Use window functions for rankings
- Write output to Parquet with partitioning
- Compare performance: Pandas vs PySpark

Project 8: Log File Analysis at Scale

- Generate or download server log files
- Parse logs with PySpark (regex)
- Extract IP addresses, timestamps, status codes
- Aggregate: Top pages, error rates, traffic patterns
- Store in partitioned Parquet format
- Optimize with broadcast joins

Project 9: PySpark ETL Pipeline

- Read from multiple sources (CSV, JSON, Parquet)

- Complex transformations with UDFs
- Join multiple large datasets
- Data quality validations
- Write to optimized format
- Performance metrics and logging

Tools to Setup:

- Install PySpark locally: <https://spark.apache.org/downloads.html>
 - Or use Databricks Community Edition (Recommended)
 - Dataset sources: <https://www.kaggle.com/datasets>
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📅 MONTH 7-8: APACHE AIRFLOW & ORCHESTRATION

Skills to Learn:

- Airflow Architecture (Scheduler, Webserver, Executor)
- DAGs (Directed Acyclic Graphs)
- Operators (PythonOperator, BashOperator, SQLOperator)
- Sensors (FileSensor, TimeSensor, HttpSensor)
- Task Dependencies
- XComs (Cross-communication)
- Scheduling (Cron expressions)
- Connection Management
- Error Handling and Retries
- Monitoring and Logging
- Dynamic DAGs
- Best Practices

Free Resources:

Learning Materials:

- Airflow Documentation: <https://airflow.apache.org/docs/>
- Astronomer Learn: <https://docs.astronomer.io/learn/>
- Awesome Apache Airflow: <https://github.com/jghoman/awesome-apache-airflow>

Video Courses:

- Marc Lamberti Airflow Course: https://www.youtube.com/watch?v=K9AnJ9_ZAXE
- Apache Airflow Complete Course: <https://www.youtube.com/watch?v=AHMm1wfGuHE>
- Airflow Tutorial for Beginners: <https://www.youtube.com/watch?v=z7xyf0lzhO8>

Installation Guide:

- Install with Docker: <https://airflow.apache.org/docs/apache-airflow/stable/start/docker.html>
- Quick Start: <https://airflow.apache.org/docs/apache-airflow/stable/start.html>

Week-by-Week Breakdown:

Week 1-2: Airflow Basics

- Learn: Airflow architecture components
- Learn: Creating first DAG
- Learn: Basic operators
- Learn: Task dependencies (set_upstream, set_downstream)
- Practice: Build simple DAGs locally

Week 3-4: Operators & Sensors

- Learn: PythonOperator, BashOperator
- Learn: SQLOperator for database operations
- Learn: Sensors for event-driven workflows
- Learn: XComs for data passing
- Practice: Build multi-step workflows

Week 5-6: Scheduling & Monitoring

- Learn: Cron expressions for scheduling
- Learn: Backfilling historical data
- Learn: Managing connections
- Learn: Error handling and retries
- Learn: Monitoring DAG runs
- Practice: Schedule production DAGs

Week 7-8: Advanced Concepts

- Learn: Dynamic DAG generation
- Learn: SubDAGs and TaskGroups
- Learn: Testing DAGs
- Learn: Best practices for production
- Practice: Build complex pipelines

Projects:

Project 10: Daily ETL Pipeline

- Task 1: Extract data from API
- Task 2: Validate data quality
- Task 3: Transform with Python/Pandas
- Task 4: Load to PostgreSQL

- Schedule: Daily at 6 AM
- Add email alerts on failure
- Use XComs for data passing
- Implement retries

Project 11: Multi-Source Data Integration

- Extract from: Database, API, CSV files
- Use Sensors to wait for files
- Parallel processing tasks
- Complex dependencies (fan-in, fan-out)
- Data validation checks
- Load to data warehouse
- Comprehensive logging

Project 12: Backfilling Pipeline

- Create DAG for historical data processing
- Implement date partitioning
- Efficient backfill strategy
- Handle late-arriving data
- Monitoring dashboard
- Document backfill process

Tools to Install:

- Docker Desktop: <https://www.docker.com/products/docker-desktop>
 - Airflow with Docker Compose
 - pgAdmin for database management
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📅 MONTH 9-10: APACHE KAFKA & STREAMING

Skills to Learn:

- Kafka Architecture (Brokers, Topics, Partitions)
- Producers (Sending messages)
- Consumers (Reading messages)
- Consumer Groups
- Kafka Connect (Source and Sink connectors)
- Serialization (Avro, JSON, Protobuf)
- Schema Registry
- Kafka Streams basics
- Offset Management
- Replication and Fault Tolerance
- Performance Tuning

Free Resources:

Learning Materials:

- Kafka Documentation: <https://kafka.apache.org/documentation/>
- Confluent Kafka Tutorials: <https://developer.confluent.io/>
- Conduktor Kafka Course (Free): <https://www.conduktor.io/kafka/kafka-tutorial>
- Apache Kafka Quickstart: <https://kafka.apache.org/quickstart>

Video Courses:

- Kafka Tutorial for Beginners: <https://www.youtube.com/watch?v=R873BINVUB4>
- Apache Kafka Crash Course: <https://www.youtube.com/watch?v=06iRM1Ghr1k>
- Kafka Fundamentals: <https://www.youtube.com/watch?v=Ch5VhJzaoal>

Practice:

- Confluent Cloud Free Tier: <https://www.confluent.io/confluent-cloud/>
- Install Kafka with Docker: <https://kafka.apache.org/quickstart>

Week-by-Week Breakdown:

Week 1-2: Kafka Fundamentals

- Learn: Kafka architecture and concepts
- Learn: Topics, partitions, brokers
- Learn: Installing Kafka with Docker
- Learn: Creating topics
- Practice: Run basic Kafka commands

Week 3-4: Producers & Consumers

- Learn: Writing Kafka producers
- Learn: Writing Kafka consumers
- Learn: Consumer groups for parallelism
- Learn: Offset management
- Practice: Build producer-consumer apps

Week 5-6: Kafka Connect & Streams

- Learn: Kafka Connect architecture
- Learn: Source and sink connectors
- Learn: Kafka Streams basics
- Learn: Serialization (Avro, JSON)
- Practice: Build streaming pipelines

Week 7-8: Advanced Topics

- Learn: Schema Registry usage
- Learn: Replication and fault tolerance
- Learn: Performance tuning
- Learn: Monitoring Kafka clusters
- Practice: Production-ready configurations

Projects:

Project 13: Real-Time Data Pipeline

- Producer: Generate real-time events (IoT simulation)
- Configure Kafka topics with partitions
- Consumer: Process messages in real-time
- Store in PostgreSQL
- Monitor lag and throughput
- Error handling and retries

Project 14: Kafka Connect Pipeline

- Source Connector: Extract from PostgreSQL
- Transform in Kafka Streams
- Sink Connector: Load to another database
- Use Avro serialization
- Schema evolution handling
- Monitor connector status

Project 15: Stream Processing App

- Multiple producers (different event types)
- Consumer groups for parallel processing
- Real-time aggregations (windowing)
- Join multiple streams
- Write to time-series database
- Monitoring dashboard with Grafana

Tools to Install:

- Docker for Kafka: <https://hub.docker.com/r/confluentinc/cp-kafka/>
- Kafka Tool (GUI): <http://www.kafkatool.com/>
- Conduktor (Kafka GUI): <https://www.conduktor.io/>

☒ MONTH 11: DATA WAREHOUSING & MODELING

Skills to Learn:

- Data Warehouse Concepts (OLTP vs OLAP)
- Dimensional Modeling
- Star Schema and Snowflake Schema
- Fact Tables (Additive, Semi-additive, Non-additive)
- Dimension Tables
- Slowly Changing Dimensions (SCD Type 1, 2, 3)
- BigQuery (Google Cloud Data Warehouse)
- Redshift Basics (AWS)
- Snowflake Basics
- Columnar Storage (Parquet, ORC)
- Partitioning and Clustering
- Materialized Views
- Query Optimization

Free Resources:

Learning Materials:

- Kimball Group Resources: <https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/>
- Data Warehouse Tutorial: <https://www.guru99.com/data-warehousing.html>
- Star Schema Design: <https://www.sqlshack.com/implementing-star-schema-in-sql-server/>

BigQuery:

- BigQuery Documentation: <https://cloud.google.com/bigquery/docs>
- BigQuery Free Tier: <https://cloud.google.com/bigquery/pricing#free-tier>
- BigQuery Tutorials: <https://cloud.google.com/bigquery/docs/tutorials>
- BigQuery Sandbox (Free): <https://cloud.google.com/bigquery/docs/sandbox>

Video Courses:

- Data Warehouse Tutorial: <https://www.youtube.com/watch?v=1D8H4f6T40o>
- BigQuery Full Course: <https://www.youtube.com/watch?v=IZKdmbM0f6l>
- Star Schema Design: https://www.youtube.com/watch?v=AHR_7jFCMeY
- Slowly Changing Dimensions: <https://www.youtube.com/watch?v=1D8H4f6T40o>

Snowflake:

- Snowflake University (Free): <https://learn.snowflake.com/>
- Snowflake Free Trial: <https://signup.snowflake.com/>

Week-by-Week Breakdown:

Week 1: Data Warehouse Concepts

- Learn: OLTP vs OLAP differences
- Learn: Data warehouse architecture
- Learn: ETL vs ELT
- Learn: Dimensional modeling principles
- Practice: Design conceptual models

Week 2: Star Schema & Snowflake

- Learn: Fact tables design
- Learn: Dimension tables design
- Learn: Star schema patterns
- Learn: Snowflake schema (normalized dimensions)
- Practice: Design schemas for use cases

Week 3: Slowly Changing Dimensions

- Learn: SCD Type 1 (overwrite)
- Learn: SCD Type 2 (add new row with versioning)
- Learn: SCD Type 3 (add new column)
- Practice: Implement SCD in database

Week 4: BigQuery Implementation

- Learn: BigQuery architecture
- Learn: Creating datasets and tables
- Learn: Partitioning and clustering
- Learn: Writing optimized queries
- Practice: Build data warehouse in BigQuery

Projects:

Project 16: E-Commerce Data Warehouse

- Design dimensional model:
 - Fact: Sales, Orders, Inventory
 - Dimensions: Customer, Product, Date, Store
- Implement in BigQuery
- Create SCD Type 2 for customer dimension
- Build ETL pipeline to load data
- Create aggregate tables
- Write analytical queries
- Optimize with partitioning/clustering

Project 17: Healthcare Data Warehouse

- Dimensional model for patient records
- Multiple fact tables (appointments, prescriptions, labs)
- Shared dimensions (patient, doctor, facility, date)
- Implement slowly changing dimensions
- ETL from source systems
- Incremental load strategy
- Summary tables and views

Project 18: Retail Analytics Warehouse

- Sales transactions fact table
- Product, store, customer, time dimensions
- Bridge tables for many-to-many
- Accumulating snapshot fact
- Partitioning by date
- Clustering by store/product
- Performance benchmarking

Tools to Setup:

- BigQuery Sandbox (Free): <https://cloud.google.com/bigquery/docs/sandbox>
 - DBT (Data Build Tool): <https://www.getdbt.com/>
 - ERD Tools: <https://app.diagrams.net/>
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📌 MONTH 12: DOCKER, TESTING & PORTFOLIO

Skills to Learn:

- Docker Basics (Images, Containers)
- Dockerfile Creation
- Docker Compose (Multi-container apps)
- Containerizing Data Pipelines
- Volumes and Networking
- Data Quality (Great Expectations)
- Unit Testing (pytest)
- Integration Testing
- Logging Best Practices
- Git Advanced
- CI/CD Basics
- Portfolio Building

Free Resources:

Docker:

- Docker Documentation: <https://docs.docker.com/>
- Docker Tutorial: <https://www.youtube.com/watch?v=fqMOX6JJhGo>
- Docker for Data Engineering: <https://www.youtube.com/watch?v=ij1N2zqPpGk>
- Play with Docker: <https://labs.play-with-docker.com/>
- Docker Curriculum: <https://docker-curriculum.com/>

Testing & Quality:

- Great Expectations Docs: <https://docs.greatexpectations.io/>
- pytest Documentation: <https://docs.pytest.org/>
- Testing Tutorial: <https://www.youtube.com/watch?v=l9fJm6GFgp4>

Git & GitHub:

- Git Documentation: <https://git-scm.com/doc>
- GitHub Skills: <https://skills.github.com/>
- Git Tutorial: <https://www.youtube.com/watch?v=RG0j5yH7evk>

Week-by-Week Breakdown:

Week 1-2: Docker

- Learn: Docker images and containers
- Learn: Writing Dockerfiles
- Learn: Docker Compose for multi-container
- Learn: Volumes for data persistence
- Practice: Containerize previous projects

Week 3: Testing

- Learn: Great Expectations framework
- Learn: pytest for Python
- Learn: Data quality checks
- Practice: Add tests to pipelines

Week 4: Portfolio Building

- Polish all 21 projects
- Write professional READMEs
- Create portfolio website
- Prepare for interviews

Projects:

Project 19: Dockerized Data Pipeline

- Containerize Airflow project
- Multi-container: Airflow, PostgreSQL, Redis
- Docker Compose configuration
- Environment variables
- Persistent volumes
- Documentation

Project 20: Data Quality Framework

- Implement Great Expectations
- Create expectation suites
- Validate data at ingestion
- Generate quality reports
- Alert on failures
- Document data contracts

Project 21: Complete Data Platform

- Combine everything learned
- Data ingestion: Kafka or batch
- Processing: Spark
- Orchestration: Airflow
- Storage: Data warehouse
- Quality: Great Expectations
- Containerization: Docker
- Architecture diagram
- Complete documentation

Tools to Install:

- Docker Desktop: <https://www.docker.com/products/docker-desktop>
- GitHub Desktop: <https://desktop.github.com/>

☒ COMPLETE SKILLS SUMMARY

After 12 months, you will have mastered:

Core Skills:

- SQL (Expert level - complex queries, optimization)
- Python (Advanced - Pandas, NumPy, data manipulation)
- Apache Spark/PySpark (Proficient - big data processing)
- Apache Airflow (Expert - workflow orchestration)
- Apache Kafka (Proficient - real-time streaming)
- Data Warehousing (Expert - dimensional modeling)

- Docker (Proficient - containerization)
- Git (Proficient - version control)

Total Projects: 21 production-grade data engineering projects

☒ ALL RESOURCE LINKS ORGANIZED

SQL Resources

- Mode SQL Tutorial: <https://mode.com/sql-tutorial/>
- SQLZoo: <https://sqlzoo.net/>
- W3Schools SQL: <https://www.w3schools.com/sql/>
- PostgreSQL Tutorial: <https://www.postgresqltutorial.com/>
- LeetCode SQL: <https://leetcode.com/problemset/database/>
- HackerRank SQL: <https://www.hackerrank.com/domains/sql>
- Use The Index, Luke: <https://use-the-index-luke.com/>

Python Resources

- Python.org Tutorial: <https://docs.python.org/3/tutorial/>
- Learn Python: <https://www.learnpython.org/>
- Real Python: <https://realpython.com/>
- Kaggle Python: <https://www.kaggle.com/learn/python>
- Kaggle Pandas: <https://www.kaggle.com/learn/pandas>
- 100 Pandas Puzzles: <https://github.com/ajcr/100-pandas-puzzles>
- Pandas Exercises: https://github.com/guipsamora/pandas_exercises

Spark Resources

- Spark Documentation: <https://spark.apache.org/docs/latest/>
- Spark by Examples: <https://sparkbyexamples.com/>
- Learning Spark 2.0 PDF: <https://pages.databricks.com/rs/094-YMS-629/images/LearningSpark2.0.pdf>
- Databricks Free Edition: <https://community.cloud.databricks.com/>

Airflow Resources

- Airflow Documentation: <https://airflow.apache.org/docs/>
- Astronomer Learn: <https://docs.astronomer.io/learn/>
- Awesome Airflow: <https://github.com/jghoman/awesome-apache-airflow>

Kafka Resources

- Kafka Documentation: <https://kafka.apache.org/documentation/>

- Confluent Tutorials: <https://developer.confluent.io/>
- Conduktor Course: <https://www.conduktor.io/kafka/kafka-tutorial>

Data Warehousing

- Kimball Group: <https://www.kimballgroup.com/>
- BigQuery Docs: <https://cloud.google.com/bigquery/docs>
- Snowflake University: <https://learn.snowflake.com/>

Docker & Testing

- Docker Docs: <https://docs.docker.com/>
- Great Expectations: <https://docs.greatexpectations.io/>
- pytest Docs: <https://docs.pytest.org/>

Video Channels

- Alex The Analyst: <https://www.youtube.com/@AlexTheAnalyst>
- freeCodeCamp: <https://www.youtube.com/@freecodecamp>
- Programming with Mosh: <https://www.youtube.com/c/programmingwithmosh>
- Corey Schafer: <https://www.youtube.com/@coreyms>
- Keith Galli: <https://www.youtube.com/@KeithGalli>

Practice Platforms

- LeetCode: <https://leetcode.com/>
- HackerRank: <https://www.hackerrank.com/>
- Kaggle: <https://www.kaggle.com/>
- Codewars: <https://www.codewars.com/>

Communities

- r/dataengineering: <https://www.reddit.com/r/dataengineering/>
- Data Engineering Discord: <https://discord.gg/2pER6dq>
- Stack Overflow: <https://stackoverflow.com/questions/tagged/data-engineering>

📅 WEEKLY SCHEDULE

Weekday (Mon-Fri): 5 hours/day

- Morning (7-9 AM): Learn new concepts (2 hours)
- Evening (7-10 PM): Practice + Projects (3 hours)

Weekend (Sat-Sun): 15 hours

- **Saturday:** Project work (8 hours)
- **Sunday:** Projects + Documentation (7 hours)

Weekly Total: 40 hours

- Learning: 10 hours
 - Practice: 10 hours
 - Projects: 20 hours
-

📅 MONTHLY CHECKPOINTS

Month 1-2: 📌 100+ SQL problems solved 📌 3 SQL projects on GitHub 📌 SQL expert level

Month 3-4: 📌 Python proficient 📌 Pandas expert 📌 3 ETL projects built

Month 5-6: 📌 Spark proficient 📌 PySpark expert 📌 3 big data projects

Month 7-8: 📌 Airflow expert 📌 Workflow orchestration mastered 📌 3 pipeline projects

Month 9-10: 📌 Kafka proficient 📌 Streaming pipelines built 📌 3 real-time projects

Month 11: 📌 Data warehousing expert 📌 Dimensional modeling mastered 📌 3 warehouse projects

Month 12: 📌 Docker proficient 📌 Testing implemented 📌 21 projects polished 📌 Portfolio complete
📌 Interview-ready

📅 EXPECTED OUTCOME

After 12 Months:

- **Role:** Data Engineer
- **Salary:** ₹12-18 LPA
- **Skills:** Expert in entire DE stack
- **Portfolio:** 21 production-ready projects
- **GitHub:** Professional presence
- **Interviews:** 90%+ success rate

Career Path:

- Year 1: Data Engineer (₹12-18 LPA)

- Year 2: Senior Data Engineer (₹18-25 LPA)
 - Year 3: Lead Data Engineer (₹25-35 LPA)
 - Year 5: Principal Engineer (₹40-60 LPA)
-

📌 START TODAY

Week 1 Action Plan:

1. Install PostgreSQL
2. Complete Mode SQL Tutorial (Introduction)
3. Solve 5 LeetCode easy problems
4. Join r/dataengineering
5. Set up GitHub account

Remember: Consistency beats intensity. 3-4 hours daily for 12 months will transform your career.

Good luck on your Data Engineering journey! 📌

END OF ROADMAP

Copy this entire text and save it. All links are clickable and free. Follow month-by-month for guaranteed success.